Efficient Programming

Today's goals

Module 1
Finding Performance Bottlenecks

Module 2
Simple Optimization Techniques

Module 3
Computer Architecture and Vectorization

Computer Architecture

Computer Architecture
Video • 6 min

Vectorization and Optimization Techniques

Maximizing Performance with Vectorization
Video • 7 min

Preparing your Application Data for Vectorization - Data Alignment
Video • 5 min

OpenMP - SIMD
Video • 4 min

Quiz

Vectorization and Parallelization in Dot Product Computation
Programming Assignment • 30 min

Module Quiz
Graded Assignment • 15 min

Module 4
Computer Architecture

Module 5
Parallel and High Throughput Computing

Programming assignment

Vectorization and Parallelization in Dot Product Computation

Launch lab



Instructions | My submissions | Discussions

What to expect

Due Sep 18, 11:59 PM IST

Unlimited attempts

In this assignment, you will optimize the performance of a dot product computation by leveraging vectorization and parallelization techniques using OpenMP. The provided code implements a dot product operation between two arrays and measures the execution time for both the baseline implementation and the vectorized, parallelized version.

Part 1: Baseline Implementation

1. Compile the provided vectorize.cpp file using the following command:

g++ -o vectorize vectorize.cpp

1. Run the compiled executable vectorize and observe the execution time for the unaligned dot product implementation.

Part 2: Vectorization and Parallelization:

In this part, you will vectorize the dot product loop using OpenMP SIMD directives and parallelize the computation across multiple threads.

1. Implement the vectorizedDotProduct function in vectorize.cpp by filling in the TODO section. This function should perform the dot product operation using OpenMP SIMD vectorization and parallelization. A. Use the #pragma omp simd directive to enable vectorization of the loop.B. Include the reduction(+:result_vectorized) clause to handle the reduction operation correctly when parallelized.

2. Compile the modified vectorize.cpp file with OpenMP support using the following command: g++ -fopenmp -O3 -o vectorize vectorize.cpp

3. Run the compiled executable vectorize and observe the execution time for the vectorized and parallelized dot product implementation.

Submission:

vectorize : The compiled executable, with the implementation of the vectorizedDotProduct function.

Report an issue

Go to next item